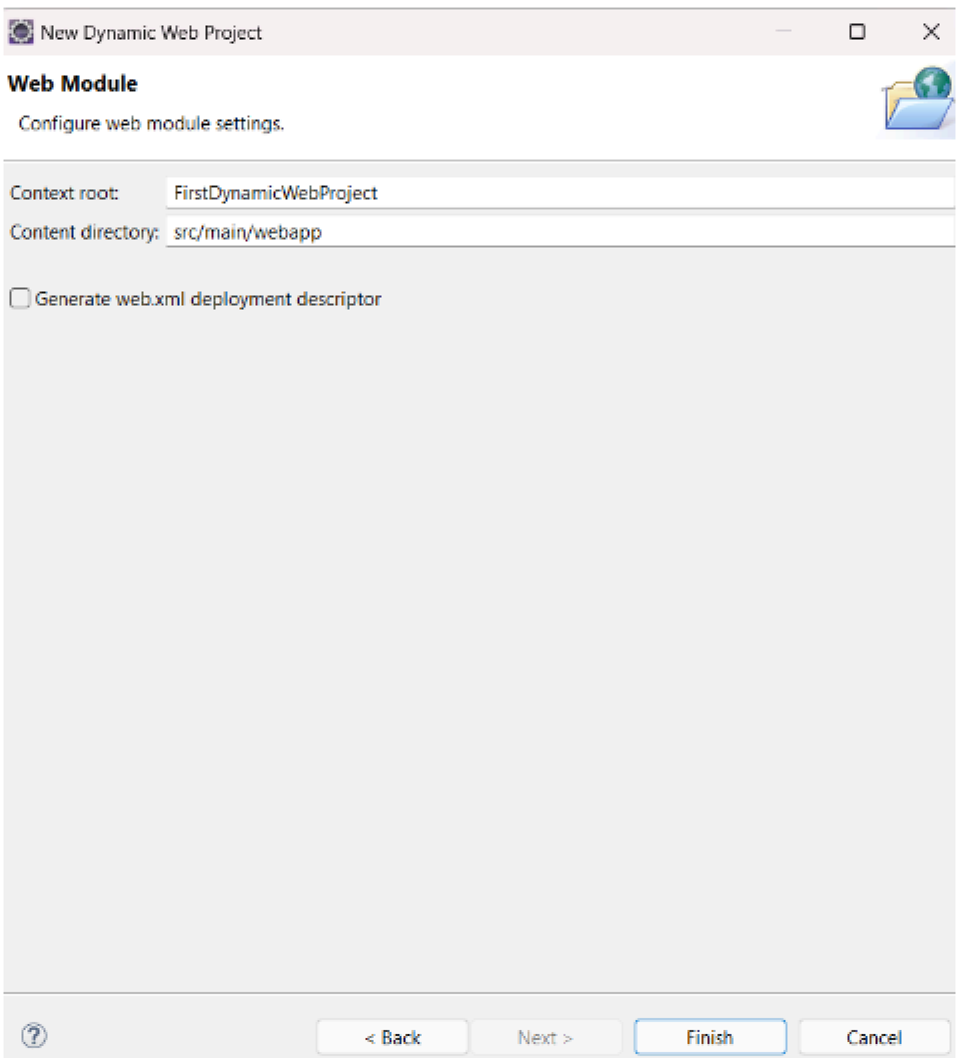
First Servlet WebApp with Dynamic Response

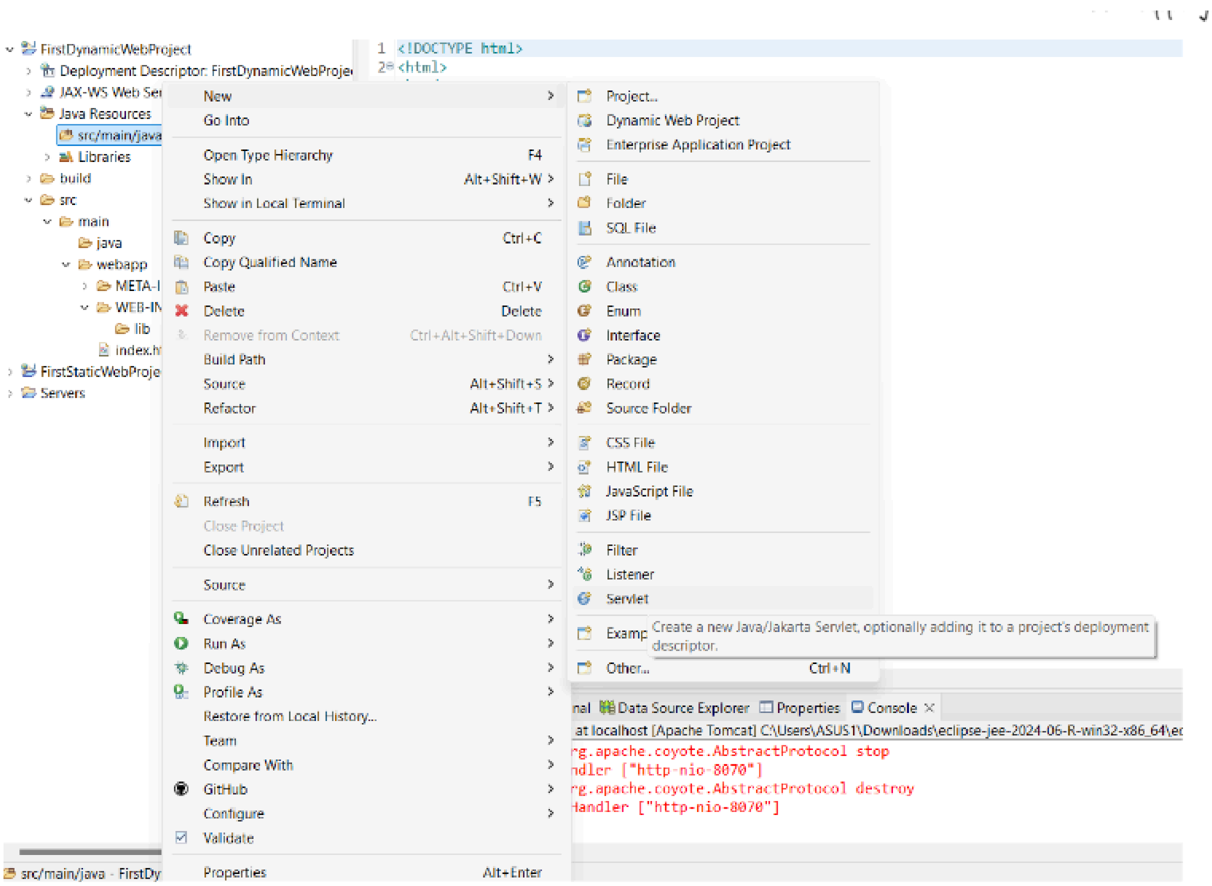
Note: If you do not stop the server and try to run another application, it will show an error that the port is busy.

👉Create First WebApp with Servlet

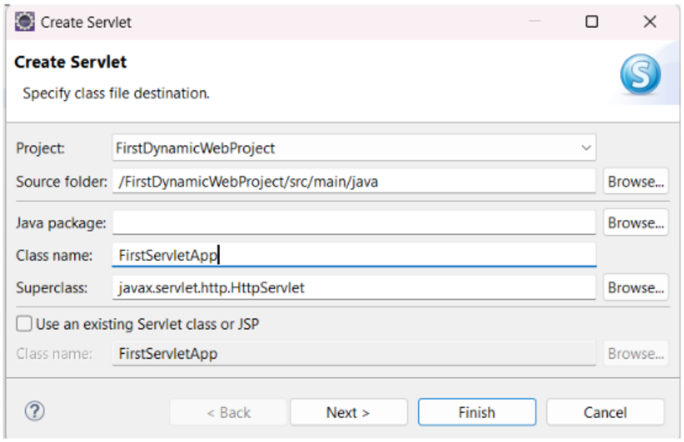
Step 1: Create a web app project without *web.xml*.



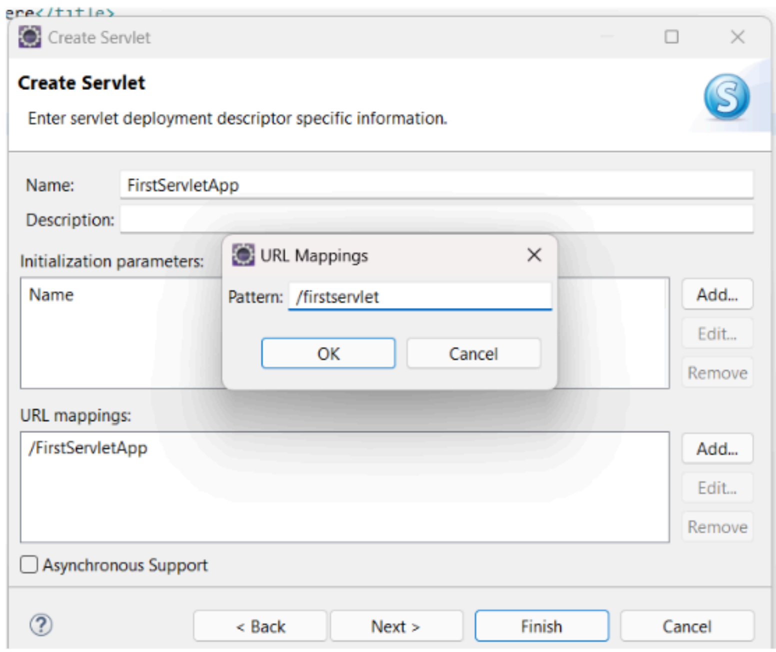
Step 2: In src/main/java, create a servlet inside a package or in the default package.



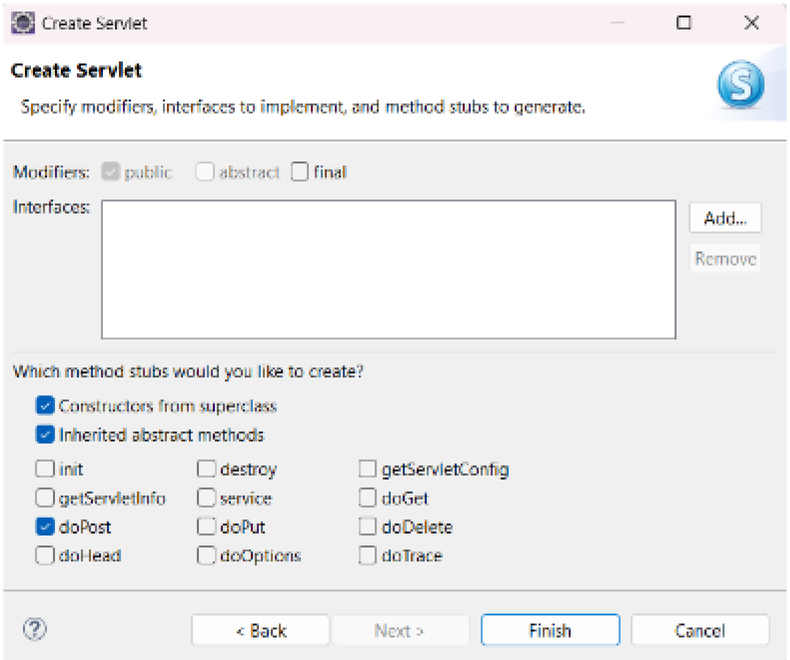
- Add class name



- Add routing path

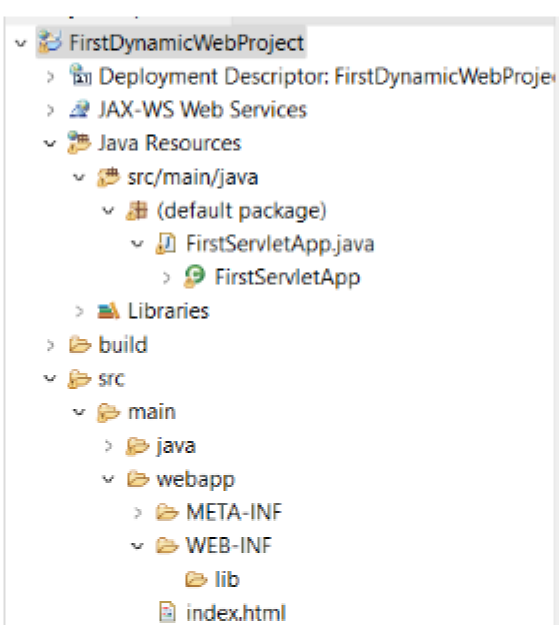


- For the first servlet project, select only this option.



Make finish.

👉Project Structure:



Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body style="font-family: Arial, sans-serif; background-color: #f4f4f9; display: flex; justify-content: center; align-items: center; height: 100vh; margin: 0;">
  <form style="background-color: #ffffff; padding: 20px; border-radius: 8px; box-shadow: 0 0 10px rgba(0, 0, 0, 0.1); width: 300px;"  action="./firstservlet" method="post">
      <h2 style="text-align: center; color: #333333;">User Information</h2>

      <div style="margin-bottom: 15px;">
          <label for="uname" style="display: block; font-weight: bold; margin-bottom: 5px; color: #555555;">Name:</label>
          <input type="text" id="uname" name="uname" style="width: 100%; padding: 10px; border: 1px solid #cccccc; border-radius: 4px;">
      </div>
      <div style="margin-bottom: 20px;">
          <label for="ucity" style="display: block; font-weight: bold; margin-bottom: 5px; color: #555555;">City:</label>
          <input type="text" id="ucity" name="ucity" style="width: 100%; padding: 10px; border: 1px solid #cccccc; border-radius: 4px;">
```

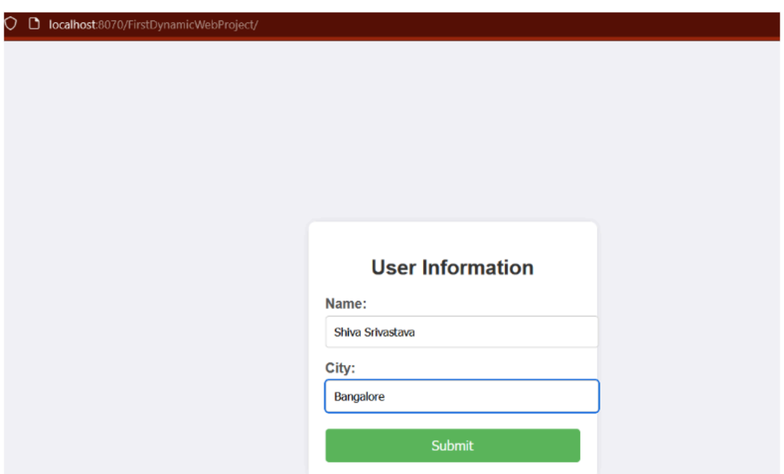
```
      </div>
      <input type="submit" value="Submit" style="width: 100%; padding: 10px; background-color: #5cb85c; color: white; border: none; border-radius: 4px; font-size: 16px; cursor: pointer;">
   </form>
</body>
</html>
```

FirstServletApp.java

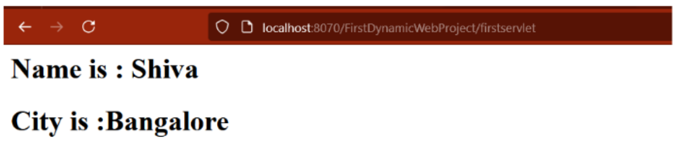
```
import java.io.IOException;
import java.io.PrintWriter;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
@WebServlet("/firstservlet")
public class FirstServletApp extends HttpServlet {

   public FirstServletApp() {
      System.out.println("Internally constructor called");
   }

        public void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException
{

                String name=request.getParameter("uname");
```

```
            String city=request.getParameter("ucity");

            PrintWriter pw=response.getWriter();
            pw.println("Name is : "+name);
            pw.println("City is : "+city);

            //use html tag for formating
//          pw.println("<h1> Name is : "+name+"</h1>");
//          pw.println("<h1> City is :"+city+"</h1>");

            pw.close();

        }
}
```

Output:



Uncomment the HTML tag.



👉Servlet Concepts and Annotations

@WebServlet Annotation

The @WebServlet annotation is used to declare a servlet.

Example:

```
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
@WebServlet("/firstservlet")
public class FirstServletApp extends HttpServlet {

}
```

Request and Response Objects

- **HttpServletRequest**: Represents the client's request. It contains information such as request parameters, headers, and attributes.
  - Methods:
    - getParameter(String name): Retrieves the value of a request parameter.
    - getParameter(String city): Retrieves the value of a request parameter.

```
String name=request.getParameter("uname");
String city=request.getParameter("ucity");
```

- **HttpServletResponse**: Represents the server's response. It allows you to set the response status, headers, and content.
  - Methods:
    - getWriter(): Returns a PrintWriter object to send character text to the client.

```
PrintWriter out = response.getWriter();
```

HttpServlet

- HttpServlet is a class provided by Java EE that extends GenericServlet and provides methods to handle HTTP requests.
- To create a servlet, extend the HttpServlet class.
- Override methods like doGet, doPost, doPut, etc., to handle specific types of HTTP requests.

```
public class FirstServletApp extends HttpServlet {
```

```
        @Override
        protected void doGet(HttpServletRequest request,
HttpServletResponse response) throws IOException {
        // Handle GET request
        }
}
```

Constructor in Servlets

- The constructor of a servlet is called when the servlet is first loaded into memory.

```
public class FirstServletApp extends HttpServlet {
        public FirstServletApp () {
        // Constructor code
        }
}
```

PrintWriter

- PrintWriter is used to send text data to the client.
- It is obtained from the HttpServletResponse object.
- Example:

```
public void doPost(HttpServletRequest request, HttpServletResponse
response) throws ServletException, IOException {
                        String name=request.getParameter("uname");
                        String city=request.getParameter("ucity");

   PrintWriter pw=response.getWriter();
```

```
  pw.println("Name is : "+name);
   pw.println("City is : "+city);

   //use html tag for formating
//       pw.println("<h1> Name is : "+name+"</h1>");
//   pw.println("<h1> City is :"+city+"</h1>");

     pw.close();

                  }
```